BASIC SERVICES · JALISCO, MEXICO · 2010–2020

Access to Basic Services and Connectivity in Jalisco, 2010–2020

Electricity, piped water, drainage and internet in homes across 125 municipalities and more than 10,000 localities, from the 2010 and 2020 censuses.



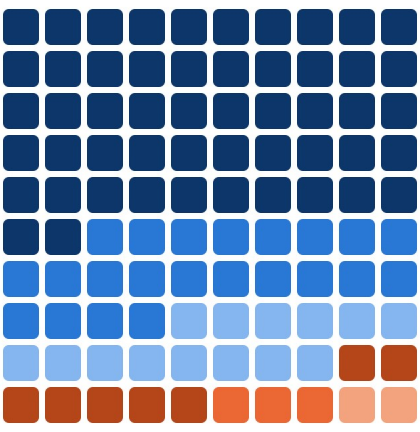
THE FIVE HOUSING SERVICES WE TRACK · COVERAGE IN 2020

Electricity	Piped water	Drainage	Non-dirt floor	Internet
99.0%	98.7%	98.6%	98.5%	61.8%

WHERE DO JALISCO'S 8.3 MILLION PEOPLE LIVE?

Most people live in cities, but most localities are tiny

Each square is 1% of the 2020 population, by size of locality.



Cities of 100,000+	52%
15,000–99,999 people	22%
2,500–14,999 people	14%
500–2,499 people (rural)	7%
100–499 people (rural)	3%
Under 100 people (rural)	2%

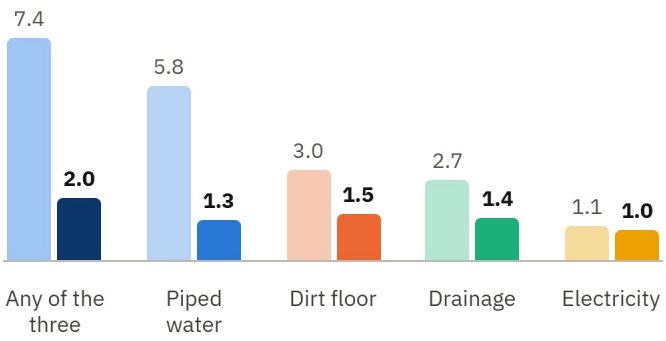
88

of every 100 residents live in urban localities, yet 8 in 10 localities have fewer than 100 people

1 · COVERAGE

Homes lacking basic services fell sharply

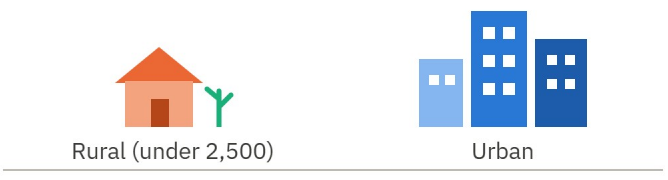
% of occupied homes lacking each service: 2010 (light) vs 2020 (solid)



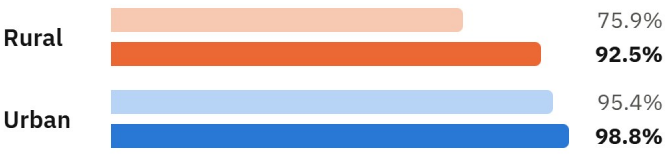
Homes without at least one of electricity, water or drainage fell from **7.4% to 2.0%**. Piped water drove most of the gain: the share lacking it fell by three-quarters.

2 · RURAL VS URBAN

The rural gap in basic services nearly closed



Homes with electricity, water and drainage: 2010 (light) vs 2020 (solid)



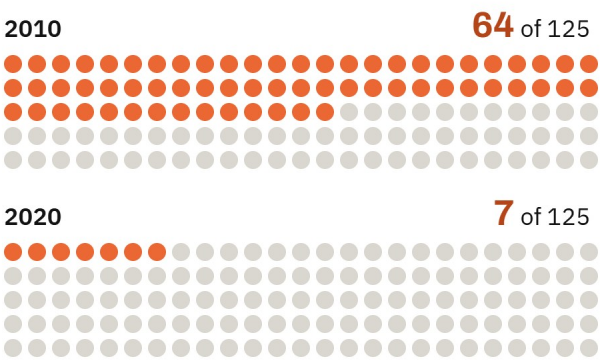
All three	Piped water	Drainage
19.5 → 6.3 pp	13.0 → 2.7 pp	10.4 → 4.1 pp

Urban–rural gap, 2010 → 2020

3 · CONVERGENCE

Lagging municipalities caught up

Each dot is one of Jalisco's 125 municipalities. **Orange** = under 90% of homes with all three services.

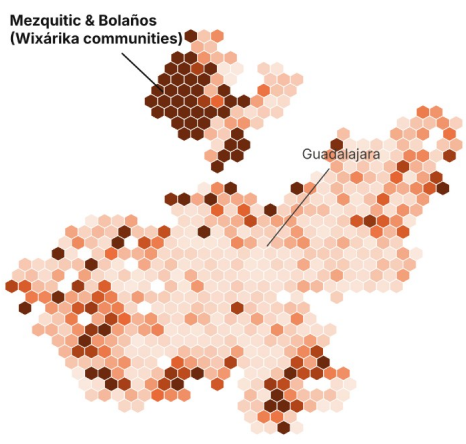


ρ = −0.97 The further behind a municipality was in 2010, the more coverage it gained.

4 · WHERE THE LAG REMAINS

The deficit is concentrated in the Wixárika north

% of homes lacking electricity, water or drainage, 2020 (localities grouped into hexagons)



COVERAGE, 2020

Mezquitic	37.2%
Bolaños	59.7%
Chimaltitán	77.5%
Jalisco	98.0%

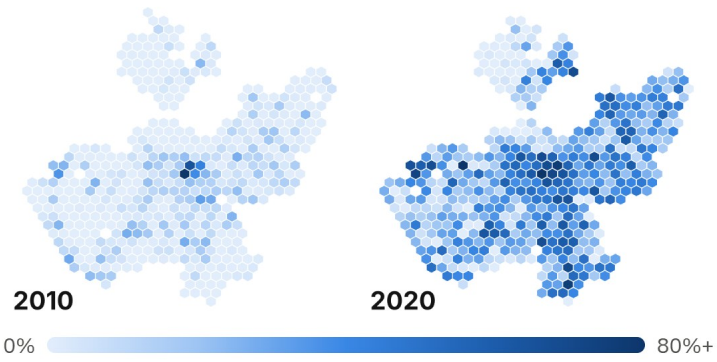
0% 40%+

In Mezquitic, 78.6% of residents speak an Indigenous language and 42.3% of homes have dirt floors (Jalisco: 1.5%).

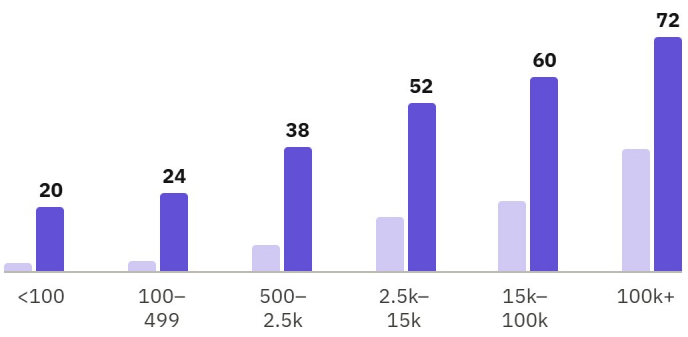
5 · CONNECTIVITY

Internet more than doubled, but the rural gap widened

% of homes with internet, by area



% with internet by locality size: 2010 (light) vs 2020 (solid)



Jalisco	Urban–rural gap
27% → 62%	24.8 → 34.2 pp

THE TAKEAWAY

Basic services are now nearly universal in Jalisco; the remaining deficit is concentrated in a few northern municipalities. Connectivity is the new divide.

Directing investment to the Northern Region and expanding rural internet access would address the largest gaps that remain.

Access to Basic Services and Connectivity in Jalisco, 2010–2020

Executive Summary

This project uses data from INEGI's 2010 and 2020 Population and Housing Censuses. Two census years were selected to add a time dimension to the analysis, in line with the project's goal of observing how access to basic services changed over a decade.

The initial exploration was challenging. The datasets are large and follow INEGI's own structure, in which state and municipal totals are stacked in the same table as individual localities, and confidential values are hidden behind special symbols. To make the information manageable, we moved from the general to the particular: we focused on the state of Jalisco and, from the more than 200 variables available, selected the 15 most relevant to the problem.

The selection was motivated by a current concern: the water supply situation in Guadalajara this year, being more specific this las months. With the 2010 and 2020 data, we can examine how basic services such as piped water, drainage and electricity were distributed across the state, and how location — the size of a locality, its municipality and its region — affects their availability.

Between 2010 and 2020, the share of homes in Jalisco with electricity, piped water and drainage rose from 92.6% to 98.0%, and the remaining deficit became concentrated in a handful of northern municipalities. Over the same decade, internet access jumped from 27% to 62% of homes, but the gap between rural and urban areas widened. This report explores both trends with locality-level data from INEGI's 2010 and 2020 Population and Housing Censuses.

Access to basic services in the home is a core measure of well-being in Mexico. INEGI's census records, for every locality in the country, how many homes have electricity, piped water, drainage, a non-dirt floor and internet. This makes it possible to see where these services reach and where they do not, down to the smallest settlement.

Jalisco, with 8.3 million inhabitants in 2020, is a demanding case. More than half of its population lives in eight large cities, while over 10,000 localities are scattered across mountains, coast and semi-desert, and 8 in 10 of them have fewer than 100 residents. Serving that dispersed population is costly, so statewide averages can hide pockets of severe deficit.

Now, how can we navigate towards that ?

The analysis is organized around five questions:

1. **How much did coverage of electricity, piped water and drainage improve between 2010 and 2020?**
2. **Did the gap between rural and urban localities close?**
3. **Did lagging municipalities catch up with the rest?**
4. **Where does the deficit persist, and who does it affect?**
5. **Does internet access follow the same pattern as basic services?**

The data come from INEGI's open-data product Principales resultados por localidad (ITER, Main Results by Locality) for the state of Jalisco, from the 2010 and 2020 Population and Housing Censuses (INEGI open data). Each row is a geographic unit and each column an aggregated census indicator. Each file comes with a data dictionary that defines every variable.

Because the census is a complete count taken once a decade, the data are two cross-sectional snapshots, not a time series. Trends are measured as change between the two censuses, and variation is studied across space: by locality size, by municipality and on maps.

The most suitable variables were selected, given the approach, of the 200–286 columns, the analysis uses 15. The table gives each variable's mnemonic, its official name in the INEGI dictionary and an its definition.

Mnemonic	Dictionary name (Spanish)	definition
ENTIDAD, MUN, LOC	Clave de entidad / municipio / localidad	Geographic keys. MUN = 000 marks state totals and LOC = 0000 municipal totals.
LATITUD, LONGITUD	Latitud, Longitud	Location of the locality in degrees, minutes and seconds.
POBTOT	Población total	Total usual residents, including an estimate for dwellings without occupant information.
VIVPAR_HAB	Viviendas particulares habitadas	Occupied private dwellings of any type, excluding those without occupant information.
VIVPARH_CV (2020 only)	Total de viviendas particulares habitadas con características	Occupied private dwellings whose characteristics were recorded.
VPH_C_ELEC	... que disponen de energía (luz) eléctrica	Dwellings with electricity.
VPH_AGUADV	... que disponen de agua entubada en el ámbito de la vivienda	Dwellings with piped water inside the home or on the lot.
VPH_DRENAJ	... que disponen de drenaje	Dwellings with drainage: public sewer, septic tank, or a pipe to a ravine, river, lake or sea.
VPH_C_SERV	... que disponen de energía eléctrica, agua entubada de la red pública y drenaje	Dwellings with all three: electricity, piped water and drainage.

VPH_PISOTI	... con piso de tierra	Dwellings with a dirt floor.
VPH_INTER	... que disponen de Internet	Dwellings with an internet service.
P3YM_HLI	Población de 3 años y más que habla alguna lengua indígena	Residents aged 3+ who speak an Indigenous language.

Counts mostly reflect population size, so each housing count was turned into a rate: dwellings with the service divided by occupied private dwellings, times 100.

Indicator	Formula
% electricity, % piped water, % drainage, % all three services	VPH_* / denominator × 100
% dirt floor	VPH_PISOTI / denominator × 100
% internet	VPH_INTER / denominator × 100
% Indigenous-language speakers	P3YM_HLI / P_3YMAS × 100
Rural / urban	Rural if POBTOT < 2,500 (INEGI threshold)

Exploratory Data Analysis

The analysis works at three levels: the state (two values per indicator), the 125 municipalities (complete in both years) and roughly 10,000 localities (to compare places by size and to draw maps). Rates for groups of localities are weighted by the number of dwellings, because 80% of localities have fewer than 100 inhabitants and a simple average would be dominated by them. Techniques include descriptive statistics, box plots, the interquartile range (IQR) rule for outliers, Spearman rank correlation and hexagon maps.

Table 1. Coverage indicators, % of occupied private dwellings. State rate and distribution across the 125 municipalities.

Indicator	State 2010 → 2020	Municipal median 2010 → 2020	Std. dev. 2010 → 2020	Skewness 2020	IQR outliers 2020
Electricity	98.9 → 99.0	98.5 → 99.5	6.1 → 3.3	−6.8	14
Piped water	94.2 → 98.7	94.0 → 99.0	9.0 → 3.2	−6.2	10

Drainage	97.3 → 98.6	95.6 → 98.7	9.5 → 6.6	−6.5	11
All three services	92.6 → 98.0	89.8 → 97.6	12.1 → 7.2	−5.8	14
Non-dirt floor	97.0 → 98.5	96.3 → 98.5	4.6 → 4.9	−5.8	8
Internet	27.0 → 61.8	10.9 → 39.3	7.5 → 13.7	0.0	0

Two features stand out. Basic-service indicators are strongly left-skewed (skewness between −5.8 and −6.8): almost every municipality is near 100%, and a few lag far behind. Those few appear as IQR outliers, so outliers are a substantive finding here, not noise. Internet behaves differently: its distribution is roughly symmetric and centered much lower, with no outliers in 2020.

The municipal medians are lower than the state rates for every indicator. The state rate is dominated by the large, well-served metropolitan municipalities, while the median reflects the typical municipality.

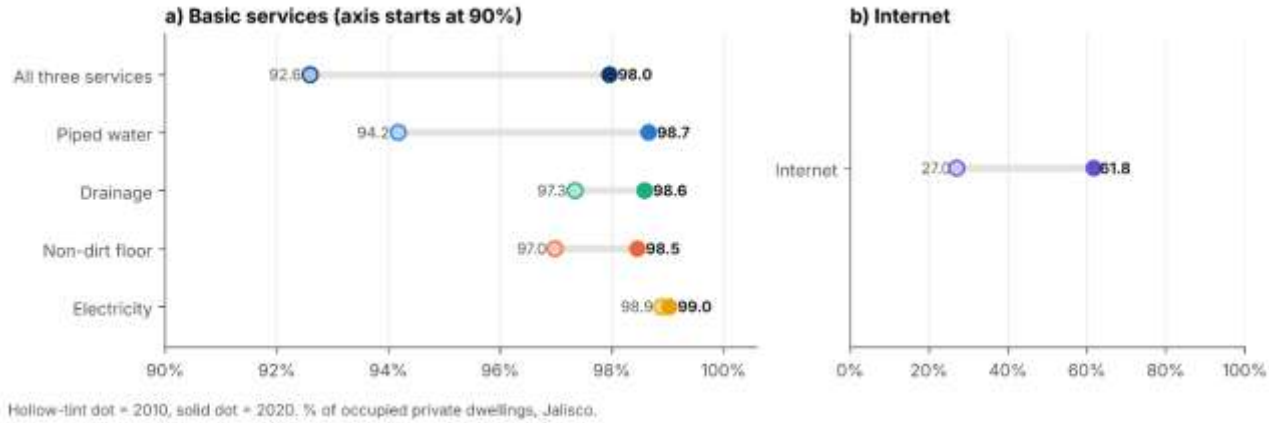


Figure 1. Statewide coverage, 2010 (light) vs 2020 (solid).

Every basic-service indicator improved. Homes with all three services rose 5.4 points (92.6% → 98.0%), and piped water drove most of that gain (+4.5 points). Electricity was already near universal in 2010 and barely moved. Internet changed on a completely different scale: +34.7 points, from 27.0% to 61.8%, more than doubling in a decade.

Panel a uses a truncated axis to make the basic-service changes visible; on a 0–100% scale they would look almost flat. That itself is a finding: by 2010 most homes already had basic services, so the decade's gains came from reaching the last 5–10% of homes.

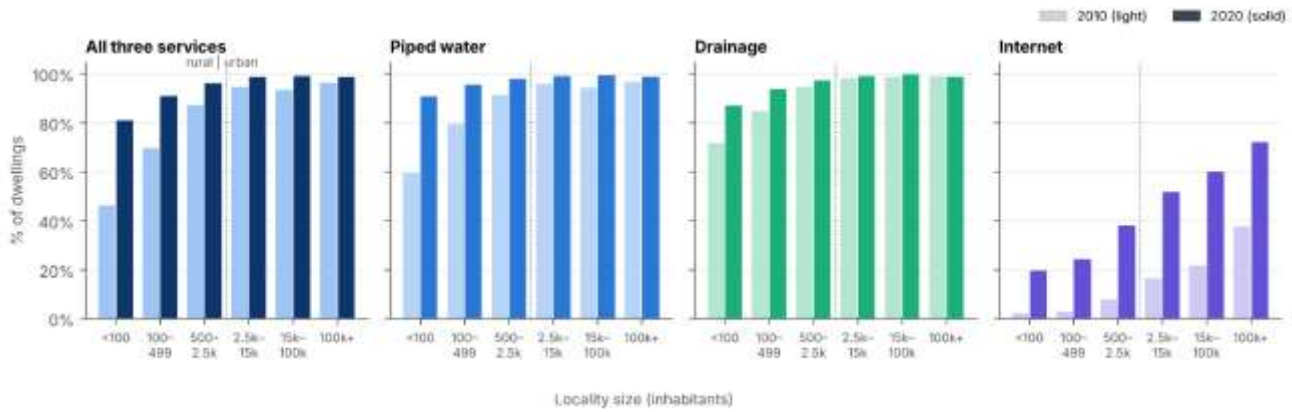


Figure 2. Coverage by locality size (inhabitants), weighted by dwellings. The dotted line separates rural (under 2,500) from urban localities.

In 2010, coverage climbed steeply with locality size. In localities with fewer than 100 inhabitants, only 46.3% of homes had all three services, compared with 96.3% in cities over 100,000. By 2020 the smallest localities had reached 81.0%, and the staircase had almost flattened above 500 inhabitants. Piped water shows the largest catch-up in the smallest places (59.4% → 90.8%).

Aggregating rural and urban localities, the gap in all three services fell from 19.5 to 6.2 percentage points; in piped water, from 13.0 to 2.8; and in drainage, from 10.5 to 4.2. Internet shows the same staircase in both years, but every step grew taller: the rural–urban gap widened from 24.9 to 34.2 points. Rural homes went from 5.6% to 31.6% with internet, urban homes from 30.4% to 65.8%.

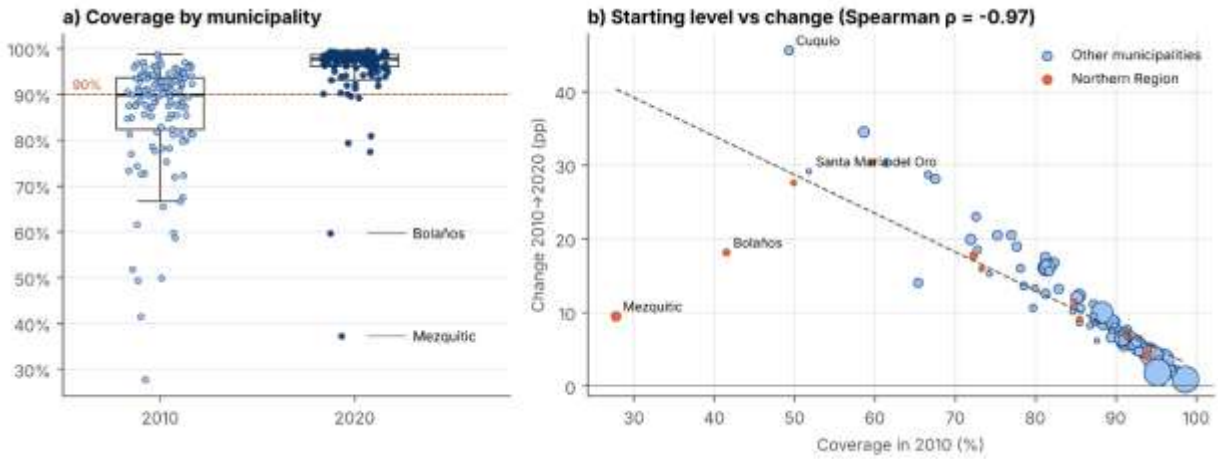


Figure 3. a) Coverage of all three services in each of the 125 municipalities. b) Coverage in 2010 against its change to 2020; point size is population, orange points are in the Northern Region.

The box plot shows the whole distribution moving up and compressing. In 2010 the middle half of municipalities ranged from 82% to 94%, and 64 municipalities were below 90%. In 2020 the box sits between 96% and 99%, and only 7 remain below 90%. The standard deviation fell from 12.1 to 7.2 points.

Panel b tests whether lagging municipalities caught up. The relationship is almost perfectly negative (Spearman ρ = −0.97): the lower a municipality's coverage in 2010, the larger its gain. Cuquío is the clearest case, going from 49.3% to 94.9% (+45.6 points). Two points break the pattern: Mezquitic (+9.5) and Bolaños (+18.1) started lowest but gained far less than the trend predicts, so they fell further behind the rest of the state.

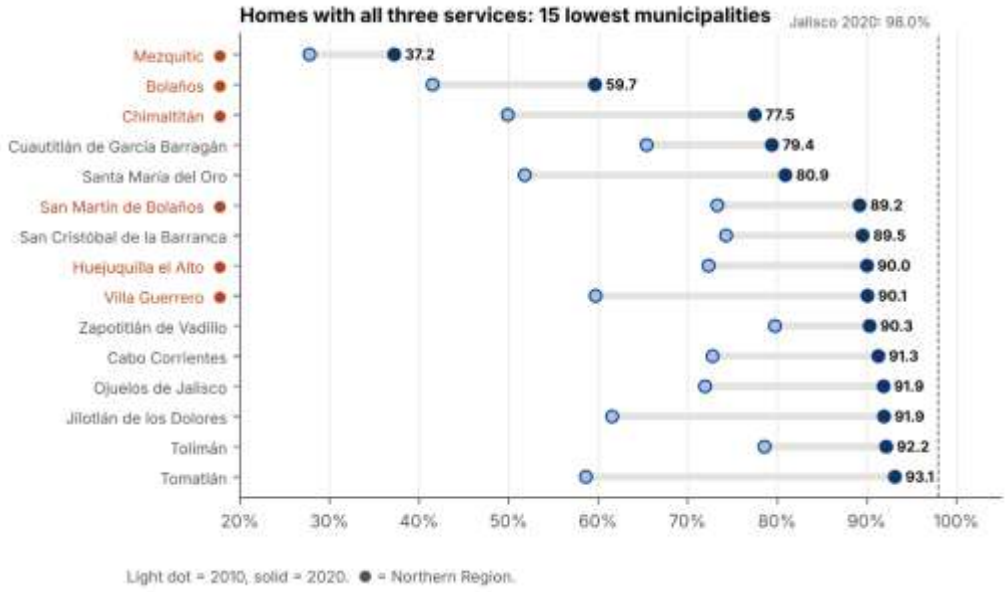


Figure 4. The 15 municipalities with the lowest coverage of all three services in 2020. Names in orange belong to the Northern Region.

Applying the IQR rule to 2020 municipal coverage, any value below 92.3% is an outlier for all three services; 14 municipalities fall below that line. The same names repeat across indicators: Mezquitic and Bolaños are the two lowest outliers in electricity, piped water, drainage and non-dirt floor. Six of the 15 lowest municipalities belong to the Northern Region, which holds 10 of Jalisco's 125 municipalities.

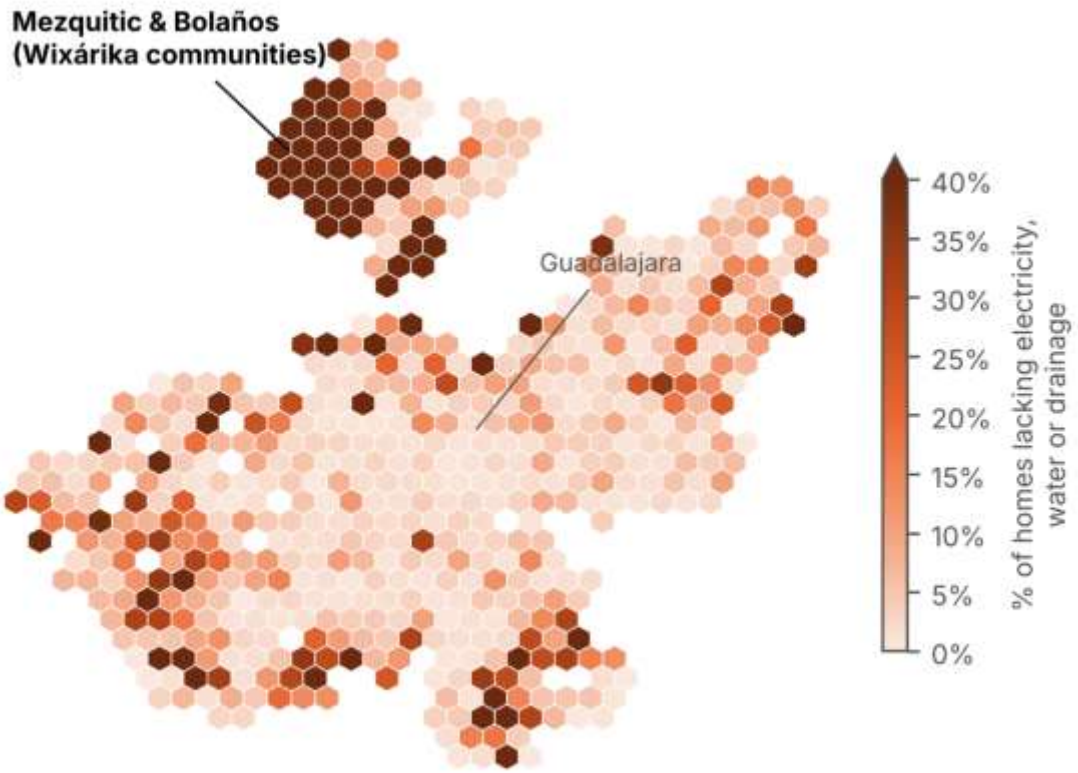


Figure 5. Share of homes lacking electricity, water or drainage, 2020. Localities are grouped into hexagons and weighted by dwellings.

The map locates the deficit. The darkest cluster lies in the far north, in the Wixárika communities of Mezquitic and Bolaños, where most hexagons exceed 40%. Smaller pockets appear along the Sierra and the southern coast, while the Guadalajara metropolitan area and the center of the state are nearly fully served.

Table 2. Spearman correlations across the 125 municipalities, 2020.

Municipal characteristic	All three services	Internet
Average years of schooling	0.67	0.66
% of population in rural localities	−0.61	−0.72
% of homes with dirt floor	−0.57	−0.53
Population size	0.40	0.65
% Indigenous-language speakers	−0.05	0.04

Coverage is higher in municipalities with more schooling and lower in more rural ones. The Indigenous-language share shows almost no correlation across all 125 municipalities, because it is close to zero in most of them. Its relevance is concentrated in the extremes: in Mezquitic, 78.6% of residents speak an Indigenous language and only 37.2% of homes have all three services.

Talking now about **connectivity**.

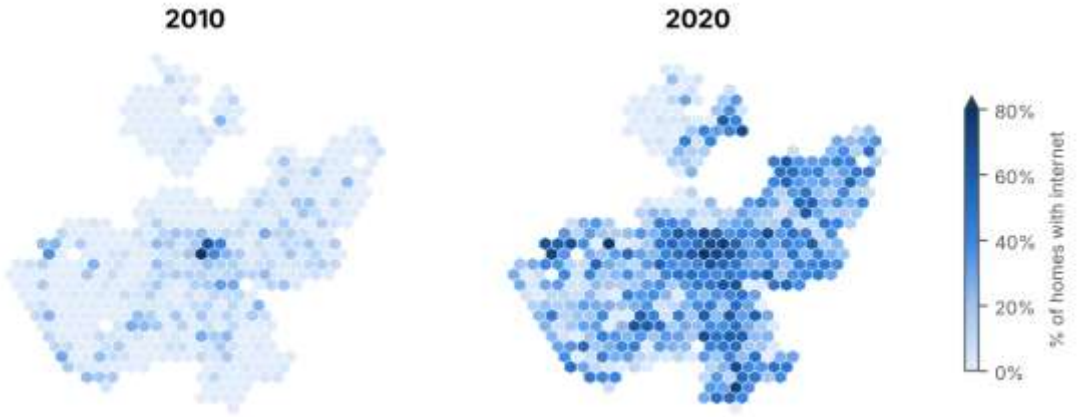


Figure 6. Share of homes with internet, 2010 and 2020, localities grouped into hexagons and weighted by dwellings.

In 2010 internet was concentrated in a few urban hexagons around Guadalajara and Puerto Vallarta. By 2020 it had spread across the center, Los Altos and the coast, while the north and the most remote rural areas stayed light.

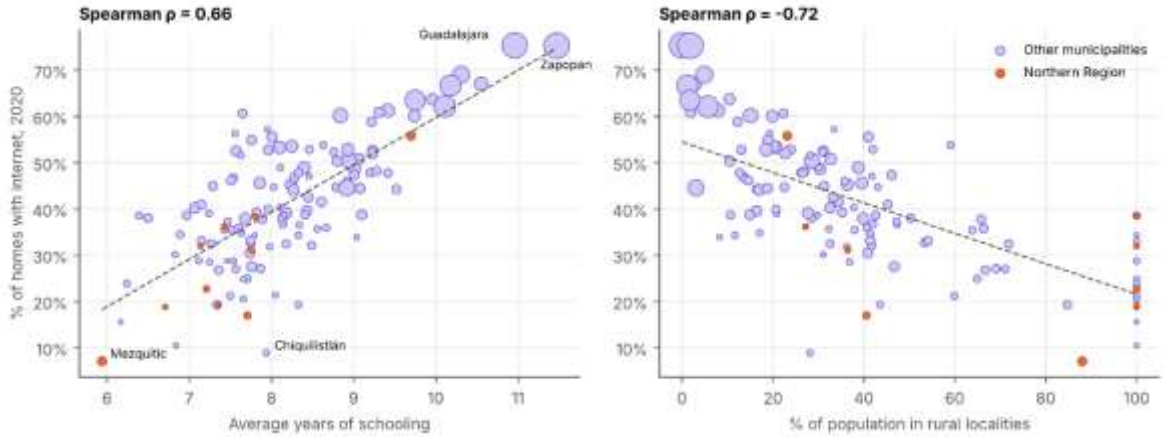


Figure 7. Internet access by municipality, 2020, against average years of schooling (left) and the share of the population living in rural localities (right). Point size is population.

Internet is more tightly linked to rurality (ρ = −0.72) and population size (ρ = 0.65) than basic services are (Table 2). Municipalities that are entirely rural cluster between 10% and 40%, regardless of other conditions. Spread across municipalities also changed: the standard deviation rose from 7.5 to 13.7 points, yet the coefficient of variation fell from 0.65 to 0.33. In absolute terms municipalities moved apart; in relative terms they moved closer. Unlike basic services, internet still has no municipality near saturation, so it remains the dimension with the most room for change.

Conclusions

Taking care of the information provided by INEGI, Jalisco nearly closed its basic-services gap between 2010 and 2020, but connectivity opened a new one. The deficit that remains in electricity, water and drainage is small and concentrated, while internet access still divides rural from urban homes.

Working with the ITER files this was found, **The files** mix state, municipal and small-locality subtotals with real localities. Separating them and checking the sum against official totals avoided counting about 24,000–28,000 people twice.

Using the wrong dwelling count in 2020 produced coverage above 100%; the right one (VIVPARH_CV) keeps every rate between 0% and 100%. With 80% of localities under 100 inhabitants and half the population in eight cities, unweighted averages describe hamlets, not people. The state rate and the municipal median also tell different stories, so both are reported. Basic-service rates are strongly left-skewed, so the outliers are the finding, not noise to remove. Two census points allow a trend but not seasonality or cycles, so locality size, municipality and maps carry most of the analysis.

So the limits on what the task need talking about. with only 2010 and originally just 2020, the analysis can measure change between censuses but cannot detect seasonality, cycles or the year-to-year path in between. Another situtation with the privacy generates that about 41% of localities have suppressed values, although they hold only 0.3% of the population. Results describe people well but say little about individual hamlets.

Finally answering the questions planted before, **how much did basic-service coverage improve?** Homes with electricity, piped water and drainage rose from 92.6% to 98.0%. Piped water drove most of the gain, while electricity was already near universal in 2010. **Did the rural–urban gap close?** Largely, yes. Rural coverage of all three services rose from 75.9% to 92.5%, and the gap with urban areas fell from 19.5 to 6.2 percentage points. **Did lagging municipalities catch up?** Yes: municipalities below 90% coverage fell from 64 to 7, and those furthest behind in 2010 gained the most (Spearman $\rho = -0.97$). **Where does the deficit persist?** In a few northern municipalities, above all Mezquitic (37.2%) and Bolaños (59.7%), home to Wixárika communities. These two also gained less than the statewide trend predicts, so their distance from the rest of Jalisco grew. And **does internet follow the same pattern?** No. Internet access rose from 27.0% to 61.8% of homes, but faster in cities, so the rural–urban gap widened from 24.9 to 34.2 points.

Appendices

The original data and the complete Python code are delivered in the archive Proyecto U1_U2.rar Apendice_ITER_Jalisco, attached to this report. Running the script on the original files reproduces every table and figure in the analysis.

Appendix A. Original datasets

All files come from INEGI's open-data portal, product *Principales resultados por localidad (ITER)*, state of Jalisco (inegi.org.mx/datosabiertos). They are used unmodified and are included in the rar.

File	Content	Size
conjunto_de_datos_iter_14CSV20.csv	ITER Jalisco, 2020 Census	10,715 rows × 286 columns
iter_14_cpv2010.csv	ITER Jalisco, 2010 Census	11,314 rows × 200 columns
diccionario_datos_iter_14CSV20.csv	Data dictionary, 2020	Definitions of every variable
fd_iter_cpv2010.csv	Data dictionary, 2010	Definitions of every variable

Appendix B. Python script

The complete analysis is in one file, analisis_servicios_basicos_jalisco.py. The same code is provided as a notebook for Google Colab: [https://drive.google.com/file/d/1FEmNMEixM7iPNtmv2OjrNULLc6HVzRlk/view?usp=sharing]
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Appendix C. Differences between the 2010 and 2020 files

The variables used are defined equivalently in both data dictionaries. The differences below were handled in the code:

- **Column names:** upper case in 2020, lower case in 2010; both were converted to lower case.
- **Dwelling types:** renamed in 2020 ("casa independiente" became "casa única en el terreno"), with the same coverage.
- **Denominator:** VIVPARH_CV (dwellings with recorded characteristics) exists only in 2020 and is the correct base for housing rates; in 2010 VIVPAR_HAB has the same scope.
- **Coordinates:** text in degrees, minutes and seconds in 2020; an unsigned integer in 2010.
- **Missing values:** N/D appears only in 2020; * (confidential) appears in both.